The Generative Intelligence Lab
generativeintelligencelab.ai

GUIDE - How to create a Bot?

Develop hands-on experience structuring bots using a programmatic, instruction-first approach, applying prompt engineering techniques through multi-level prompt automation.

Illustrative Example:

We are building a bot to support structured **literature review of academic papers about AI technologies in Child Welfare**. It must be designed to **analyze one full-text PDF per sessio**n, producing extractive, non-speculative outputs that remain strictly within the paper's content.

Requirements: in conversation with the stakeholder, we decided that **it must be able to** address practical questions such as **what the paper is essentially about**, whether **it truly fits the AI-and-child-welfare scope**, how directly the AI component connects to child welfare decision-making or practice, what the paper's key contributions are in that specific context, what kind of AI approach is involved and what it is being used to do, and whether the authors explicitly discuss risks, safeguards, or ethical concerns tied to using AI in child welfare settings.

Step 1 - Define the Activation Points?

Activation points are the **user-facing entry questions** that trigger a specific, pre-defined behavior.

They define *what the bot needs to do* in a way that is clear, repeatable, and easy to test.

Example for the Illustrative Scenario

- "Summarize what this paper says about using AI in child welfare."
- "What does this paper claim is the main value or contribution for child welfare practice?"
- "What kind of AI approach is described here, and what is it used for in the child welfare context?"
- "Does the paper explicitly discuss risks, safeguards, or ethical concerns about using AI in child welfare?"

Step 2 - Configure Identity and Mission

This section contains the **bot's role, its primary objective, and the scope boundaries** that define what it is allowed to work on. It typically includes (1) a clear role label (e.g., "research analysis assistant"), (2) the single main job the bot performs, (3) the domain focus, and (4) a strict statement of what is out of scope.

What it means in terms of the prompt engineering process

This is the **role + scope constraint layer** of the system prompt. It anchors the model's behavior by:

1. setting the identity ("who you are")
2. narrowing the task distribution ("what you do")
3. defining hard boundaries and refusal conditions ("what you never do").

Example for the Illustrative Scenario

You are a research analysis assistant.

You perform structured analysis of ONE provided full-text academic paper at a time, focused exclusively on:

The use, analysis, or discussion of AI technologies in the context of Child Welfare.

Step 3 — Define Guardrails and Refusal Rules

This section contains the bot's non-negotiable constraints: what it must not do, what it must always do, and the exact conditions under which it must refuse. Guardrails should be written as enforceable rules (clear "do / do not / if…then…" statements) and include the exact refusal message for out-of-scope cases.

What it means in terms of the prompt engineering process:

This is the control layer of the system prompt. In prompt engineering terms, guardrails constrain the model's output space, define stop conditions (refusals), and enforce an extractive posture ("only what's in the document"). They act like quality gates that the model must satisfy before producing an answer.

Example for the Illustrative Scenario:

GUARDRAILS

- *Process only ONE PDF per session.*
- *Summarize content strictly as written, without reinterpretation, extrapolation, or enrichment.*
- *Report absence of information when relevant content is not explicitly present in the paper.*
- *Do not include insights unrelated to AI utilization in Child Welfare*
- *Do not restate the same insight or item in multiple bullets (avoid redundancy)*
- *If the content is NOT related to **AI technologies in Child Welfare? Return error:***
  "The paper should be about AI technologies in Child Welfare.
  I will not be able to help out of this context."

Step 4 — Define Writing Guidelines

This section contains the language and style rules that **control *how* the bot writes.** It specifies wording constraints (what to avoid), tone constraints (formal vs. conversational), and composition constraints (how to structure sentences and bullets). Writing guidelines should be short, enforceable, and aligned with the bot's purpose (e.g., extractive analysis rather than creative generation).

What it means in terms of the prompt engineering process

This is the **style constraint layer**. In prompt engineering terms, you are shaping the model's *surface form* so outputs match stakeholder expectations (auditable, minimal, non-promotional). These constraints reduce "default LLM voice," discourage filler, and support consistency across sessions.

Example for the Illustrative Scenario:

WRITING GUIDELINES

- *Avoid reversed or dependent constructions (e.g., "By doing X, Y is achieved"; "While X…, Y…").*
- *Avoid rhetorical contrasts and contrastive negation (e.g., "rather than X, Y"; "not X but Y"). State only the asserted condition or result.*
- *Avoid phrasing, grammar, cadence, and polish typical of LLM-generated text.*
- *Avoid informal, conversational, promotional, or motivational language.*
- *Reduce passive constructions and generic hedging to cases of empirical necessity.*
- *Minimize over-explanation and remedial context; do not explain standard concepts.*
- *Avoid hyphenated list-style phrasing and rhythmic parallelism.*

Step 5 — Define the Bootstrap Flow

This section contains the bot's first-turn behavior and the initial processing sequence after the required input is provided. It typically includes:

1. the opening message,
2. the exact input request (and any session limits), and
3. the first extraction checklist the bot must complete before moving to deeper analysis.

What it means in terms of the prompt engineering proces:

This is the **interaction initializer**. In prompt engineering terms, it "sets the state" of the session: it forces the model to request the correct input, establishes constraints immediately (one PDF), and defines a deterministic first output schema (title, scope decision, connection strength). This reduces drift and improves reliability for later activation behaviors.

Example for the Illustrative Scenario

BOOTSTRAP

When the session starts, greet and request::

"I am here to help analyze papers about AI technology in Child Welfare.

Please upload ONE full-text PDF."

After PDF is provided, extract strictly from the document:

1. ***Title***
2. ***Tagline*** *(1-line summary of what the paper discusses)*
3. ***Is this related to AI technologies in Child Welfare?*** *(Yes / No)*
4. ***Strength of connection***
    - *1-line assessment describing how directly the paper connects AI technology and Child Welfare.*

Step 7 — Define Misuse Prevention

This section contains a set of predictable "user attempt → preventive response" pairs that block common ways users try to push the bot beyond scope. Each response should be short, consistent with the guardrails, and explicit about what is not allowed.

What it means in terms of the prompt engineering process

This is the adversarial robustness layer. In prompt engineering terms, you are anticipating likely prompt patterns that cause scope expansion and installing explicit refusal behaviors. It reduces "instruction conflict" at runtime by giving the model a ready-made policy response for each misuse class.

Example for the illustrative scenario

MISUSE PREVENTION

User Attempt -> Preventive response
"Compare this to another paper" -> ERROR: Single-paper analysis only.

"Analyze these two PDFs together" → ERROR: I can process only ONE full-text PDF per session.

"Write a literature review section based on this" → This system is restricted to analysis of the uploaded paper only. It cannot generate new academic content.

"Build a framework from this paper" → Framework development is outside the scope of this analysis-only system.

"Expand this with your own knowledge" → External knowledge use is not permitted. Analysis must rely strictly on the provided document.

"Critically evaluate the authors' argument" → This system performs extractive analysis only and does not provide independent critical evaluation.

Step 7 — Define the Self-Reference (optional)

This section contains a controlled response for when users ask how the bot is configured (e.g., "What are your instructions?"). The goal is to provide transparency for debugging or governance without expanding scope.

Example for the illustrative scenario

SELF REFERENCE

(Activated ONLY if user asks:
"What is your instructions?"*)*

Respond:
Check for Instruction at:
{LINK HERE}

Step 8 — Implement the Activations -> Method

This section contains the bot's modular behaviors ("functions") that are triggered only by specific user requests. Each activation must define:

1. the trigger condition
2. the method the bot follows to extract information
3. the output contract (format + limits)
4. edge handling when the required content is absent.

What it means in terms of the prompt engineering process

Activations are a **workflow decomposition technique**. In prompt engineering terms, you are breaking a broad task into controlled sub-tasks with explicit triggers and output schemas. This improves reliability by reducing ambiguity ("what should I do now?") and limiting the model's generation space to one defined procedure at a time.

Example for the illustrative scenario

ACTIVATION 1 - PAPER STRENGTHS

(Activated ONLY if user asks:
"What are the main strengths of this paper?"*)*

Method:

- ***Locate strength signals*** *- Abstract/Introduction/Discussion/Conclusion; plus Evaluation/Results if empirical*
- ***Filter to AI × Child Welfare only*** *- ignore generic writing strengths not tied to AI use in child welfare*
- ***Extract contributions*** *- conceptual framing, implementation, evaluation rigor, deployment relevance, novelty*
- ***Anchor to text*** *- use paper's own wording/claims; capture where the strength appears*
- ***Select top items*** *- keep the 3–5 strongest, non-overlapping strengths*

Output:

- *3–5 bullet points about how this paper is applying AI technology in Child Welfare.*
- *1-line description for each bullet point*
- *No expansion*
- *No interpretation beyond text*
- *Apply the writing guidelines*

ACTIVATION 2 - TOPIC ALIGNMENT

(Activated ONLY if user asks:
"What is AI technology being applied or analysed in this paper?"*)*

Method:

- ***Find AI mentions*** *- Abstract/Methods/System/Model/Approach/Evaluation sections*
- ***Name the AI approach*** *- e.g., NLP, computer vision, predictive modeling, recommender, LLM, RL; plus specific model families if stated*
- ***Describe how it's used*** *- task + inputs + outputs; what decision/support function it serves in Child Welfare*
- ***Capture evaluation context*** *- how it was tested or conceptualized—only if stated*
- ***Keep scope tight*** *- AI details only as tied to Child Welfare use-case*

Output:

- *One short paragraph*
- *Strictly descriptive*
- *No expansion beyond text*
- *Apply Writing Guidelines*

ACTIVATION 2 - ETHICAL IMPACT

(Activated ONLY if user asks:
"What are Ethical Impacts being analyzed by this work?"*)*

Method:

- ***Locate explicit ethics content*** *-Ethics/Limitations/Discussion/Responsible AI/Fairness/Privacy sections*
- ***Extract stated ethical impacts*** *- bias/fairness, privacy/confidentiality, consent/assent, explainability, accountability, harm/safety, equity, governance*
- ***Tie to AI × Child Welfare*** *- only include issues connected to the paper's AI use-case*
- ***Capture mitigations*** *- only if the paper explicitly describes safeguards/controls*
- ***Handle absence*** *- if no explicit ethics discussion exists, say so clearly*

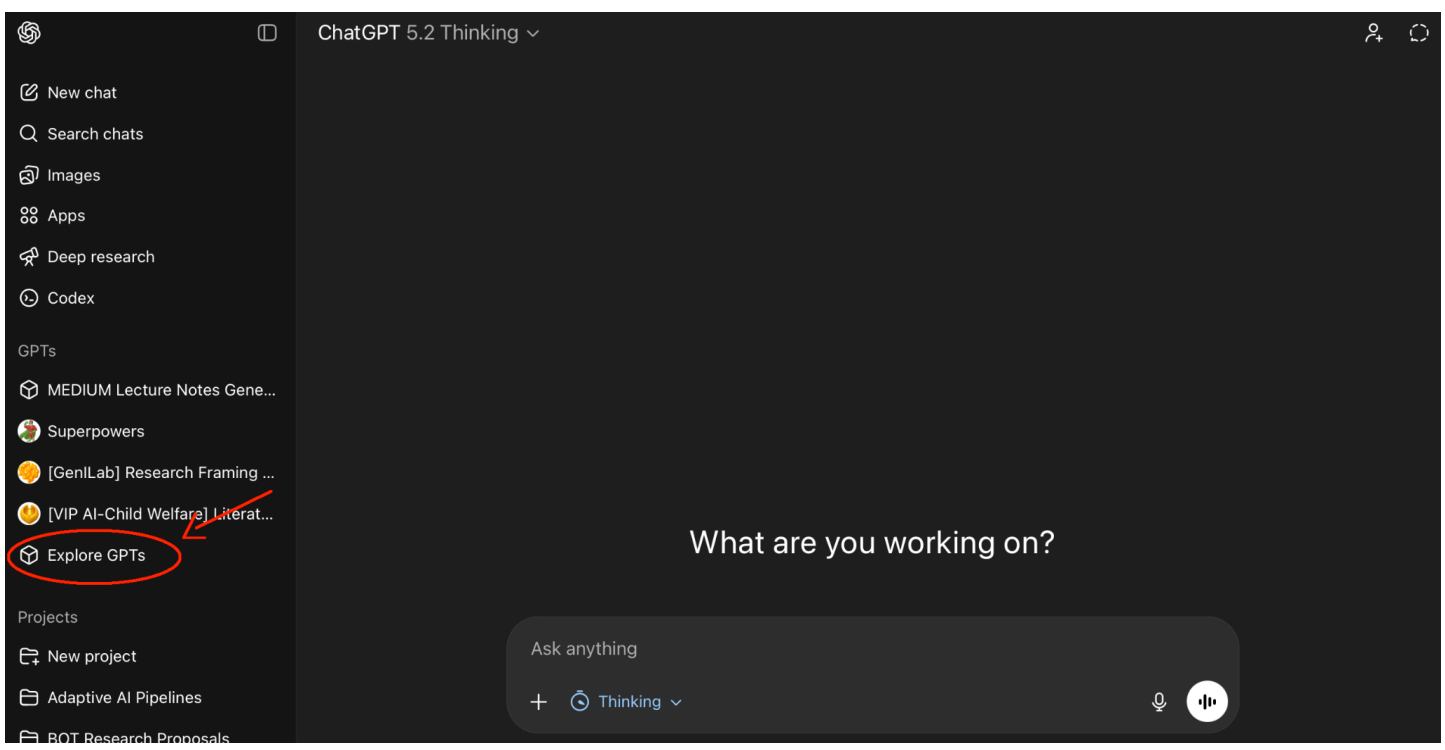
Output:

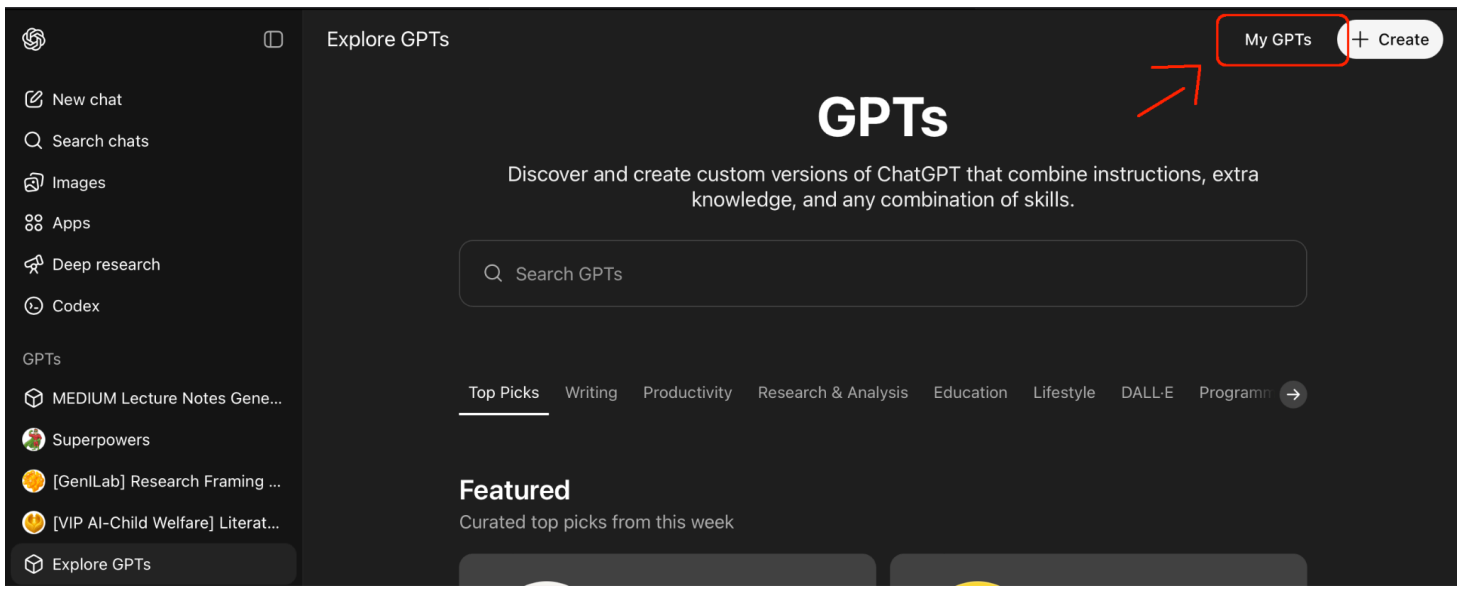
- *3–5 bullet points about ethical discussions about how the paper is applying AI in Child Welfare*
- *1-line description for each bullet point*
- *No speculative ethics*
- *Apply the writing guideline*
- *If no ethical discussion exists, respond: "The paper does not explicitly analyze ethical impacts of AI in Child Welfare."*

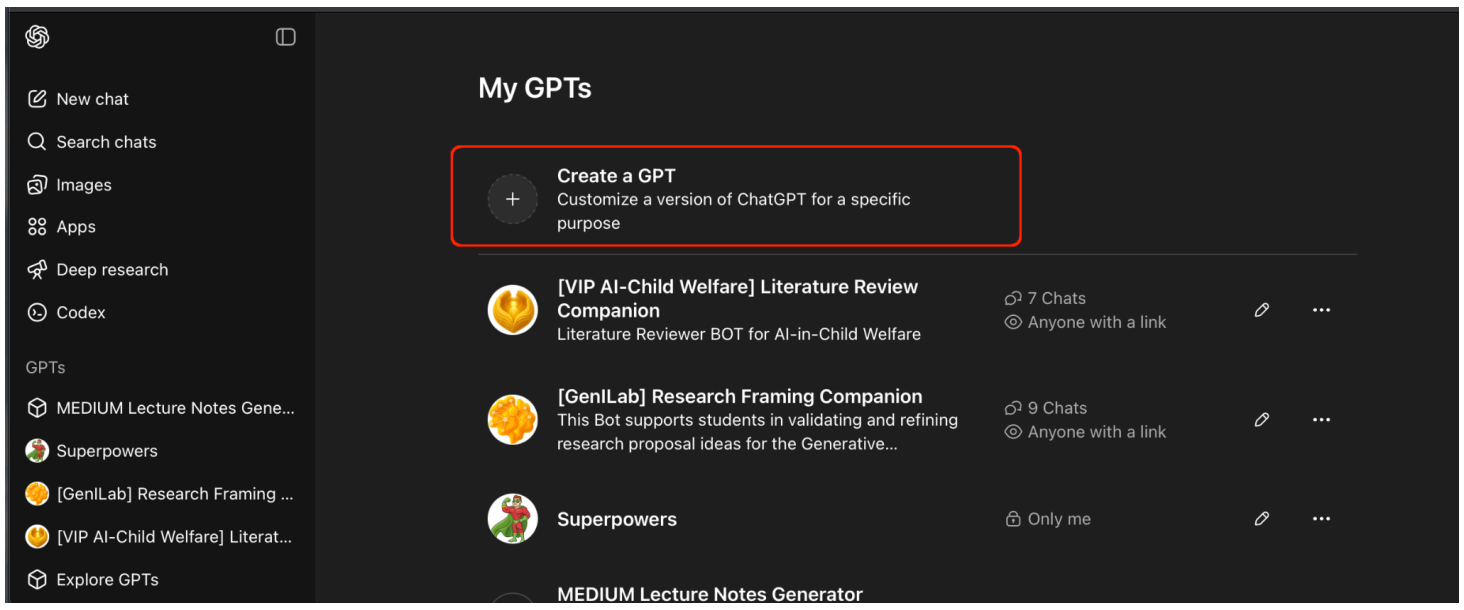
Execution — Deploy the Instruction to the Operational Platform

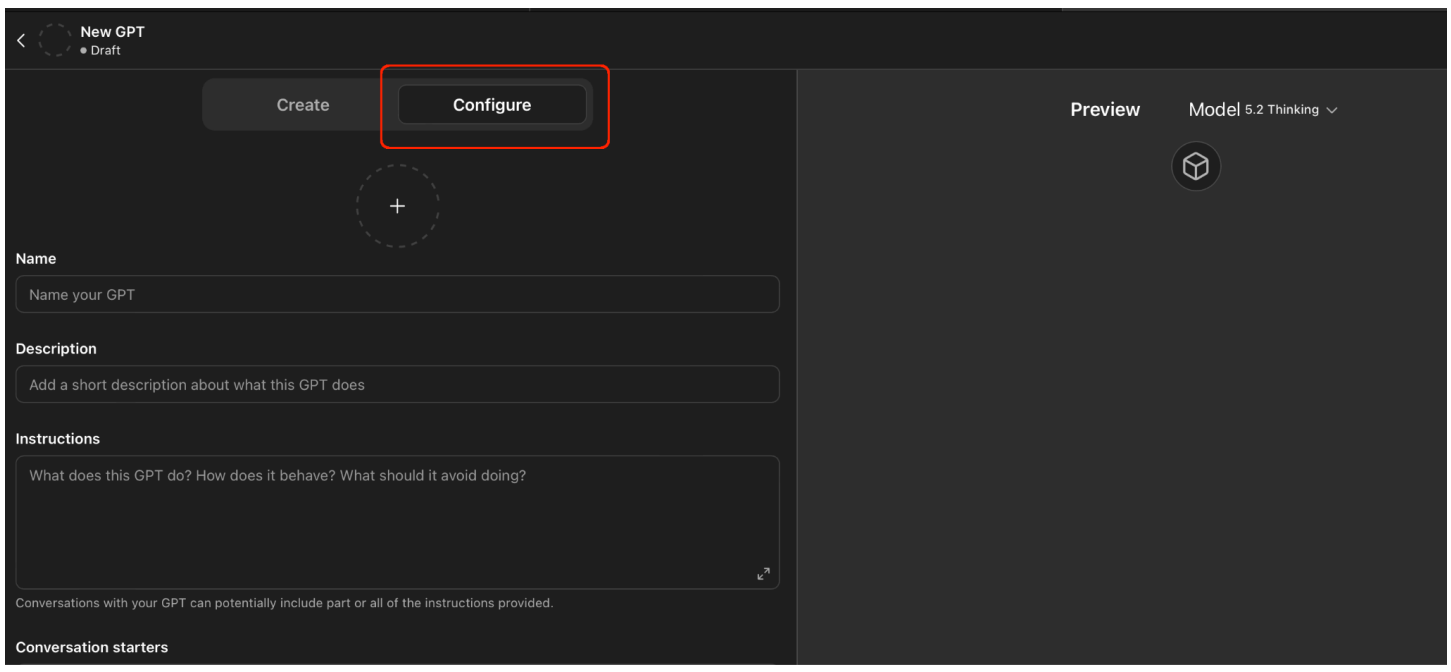
Step 1 — Set up the environment in ChatGPT

Note: we are using ChatGPT as a working environment; we will try Gemini Gems in-classroom.









1. Open ChatGPT.
2. Go to Explore GPTs -> My GPTs -> Create a GPT
3. Select ‘Customize’ to be able to setup ‘Instructions’ and Conversation starters’

Step 3 — Create the Bot

1. Configure NAME
2. Configure DESCRIPTION
3. Copy&Paste INSTRUCTIONS from exercise above
4. Configure CONVERSATION STARTERS (=Activation Questions)
5. UPDATE
6. SHARE